

SEMESTER-III

URA302: PYTHON PROGRAMMING				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: This course is designed especially to formulate and solve mathematical and mechanical problems with python programming. Furthermore, this course will also be the basis of solving various engineering problems through python.				
Syllabus Python part I: Basic data types: numbers, booleans, strings; Containers: list, slicing, loops, dictionaries, sets, tuples; Functions, Classes; NumPy: Arrays, Array indexing, Data Types, Array math, Broadcasting; SciPy: integration, interpolation, linear algebra, eigenvalues, spatial data structures; Matplotlib: Plotting, Subplots, Images Pandas: Data Manipulation, Data Cleansing and modification Python part II: 2D geometry: point, vector, lines and segments, polygons, affine transformations, Graphics and simulations: vector images, GUI and canvas, geometric primitives, simulations and time loops, affine transformations Python part III: Systems of equations: matrices and vectors, linear equations, Cholesky decomposition; Robot application: navigation and manipulation of a mobile robot using open source Python libraries				
Laboratory Work 1. Introduction to Python: Setting up a Python environment, writing and running basic Python programs, and using Jupyter Notebooks. 2. Basic data types and containers: Writing Python programs that manipulate numbers, booleans, strings, lists, dictionaries, sets, and tuples. 3. Functions and classes: Writing Python programs that define and use functions and classes to solve problems. 4. NumPy: Writing Python programs that use NumPy arrays and functions to perform numerical computations. 5. SciPy: Writing Python programs that use SciPy functions to solve scientific computing problems such as integration, interpolation, and linear algebra. 6. Matplotlib: Writing Python programs that use Matplotlib to create plots and visualizations from data. 7. Pandas: Writing Python programs that use Pandas to manipulate and clean data, and perform analysis on it. 8. 2D Geometry: Writing Python programs that use 2D geometry concepts such as points, vectors, and polygons to solve problems. 9. Graphics and Simulations: Writing Python programs that use graphics libraries such as Pygame or Tkinter to create visual simulations. 10. Robot Applications: Writing Python programs that use open-source libraries for mobile robot and manipulator applications.				

Approved in 113th meeting of the Senate held on September 7, 2024, respectively. Further, revised in 1st meeting of the Academic Council held on September 11, 2025.

Course Learning Objectives (CLO)

The students will be able to:

1. write, compile and debug programs in Python language, use different data types, operators, and functions.
2. use Python libraries to solve 2D geometry problems
3. use Python libraries to formulate and solve numerical problems.
4. address robotic applications through Python programming.

Text Books

1. Orbaiceta, A., Hardcore Programming for Mechanical Engineers, No Starch Press (2021), 1st ed.

Reference Books/Online Resources

1. https://github.com/psrana/Learn-Python-in-2-hr/blob/main/Learn_Python_in_2_hr.ipynb
2. https://numpy.org/doc/stable/user/absolute_beginners.html
3. <https://docs.scipy.org/doc/scipy/tutorial/index.html#user-guide>
4. https://pandas.pydata.org/docs/user_guide/index.html#user-guide

Evaluation Scheme:

Sr.No.	Evaluation Elements	Weightage (%)
1.	MST	25
2.	EST	40
3.	Sessionals (May include assignments/quizzes)	15
4	Laboratory evaluation	20

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